

GIOVANNI ZHANG

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EDUCATION

ETH ZURICH

Master of Science in Quantum Engineering

Zurich, CH

2023-Present

- GPA: 5.75/6
- Selection of courses include:
 - Solid State Theory, Quantum Dot Spin Qubits, Superconducting Qubits, RF Design Techniques, Quantum Information Processing, Quantum Optics

EPFL

Bachelor of Science in Physics, Exchange semester

Lausanne, CH

2022-2023

- GPA: 5.75/6

POLITECNICO DI MILANO

Bachelor of Science in Physics Engineering

Milan, IT

2020-2023

- Graduated with 110/110 cum laude

RESEARCH EXPERIENCE

QUANTUM DEVICE LAB (QUDEV), PROF. ANDREAS WALLRAFF

Zurich, CH

Research Student

09/2025-Present

- Returned to the lab as a Research Assistant and Master's Thesis student working on 3D integrated superconducting qubits
- Designing the next generation of large scale modular devices

ENGINEERING QUANTUM SYSTEMS GROUP, PROF. WILLIAM D. OLIVER

Boston, MA

Visiting Student (Master's Thesis)

09/2025-09/2025

- Year-long Master's thesis project at the Massachusetts Institute of Technology (MIT) cancelled due to VISA issues
- Originally meant to work on hybrid qubit platforms and fabrication optimization of superconducting qubits.

QUANTUM MOTION

London, UK

Quantum Hardware Intern

02/2025-Present

- Extraction of silicon-based spin qubits' parameters at cryogenic temperatures
- Perform and improve RF reflectometry readout schemes

QUANTUM DEVICE LAB (QUDEV), PROF. ANDREAS WALLRAFF

Zurich, CH

Research Student

09/2024-02/2025

- Joined the research group as a student researcher to investigate Two Level Systems (TLS) coupling with superconducting qubits.
- The project is set to build the infrastructure to study E-field based AC/DC mitigation techniques of TLS in transmon qubits.
- It includes the design of the qubit parameters and on-chip Low Pass filters using in-house libraries based on GDSpy and Ansys EDT and HFSS

QUANTUM COHERENCE LAB, PROF. DOMINIK ZUMBUHL

Basel, CH

Summer Intern

07/2024-09/2024

- Joined the group as a summer intern working on their partnership project with Intel Corporation
- Assisted in cryogenic measurements of Intel's spin-based Tunnel Falls chip
- Worked with both dry and wet dilution refrigerators, assisting in He/N transfers and replacement of electronic components.

- Developed knowledge in QCoDes, BEOL processes (glueing, wirebonding, cleaning), and tuning quantum dots.

ENSSLIN'S NANOPHYSICS GROUP, PROF. KLAUS ENSSLIN

Zurich, CH

Semester Project Student

04/2024-07/2024

- Joined the group for a semester project focused on 2 parts:
 - COMSOL simulation of a Magic Angle Twisted 4-Layers Graphene – based Josephson Junction
 - Dip stick characterization of Magic Angle Twisted Bilayer Graphene devices

AQUA LAB, PROF. EDOARDO CHARBON

Neuchatel, CH

Research Assistant

12/2022-03/2023

- Worked in the group of Edoardo, specializing in single photon detectors
- Assisted in setting up a SPADs-based sensor for road quality assessment, embedded on FPGAs.

AWARDS AND SCHOLARSHIPS

- **ETH Quantum Hackathon Winner, 2024**
 - Winners of the challenge “Quantum Neural Network for designing Differentiable Quantum Circuits to solve PDEs using PyTorch and Qadence” proposed by PASQAL
- **“Ermenegildo Zegna Founder’s Scholarship”, 2023**
 - National scholarship for 60 recipients consisting of 15'000 CHF/year
- **“Excellence fellowship” recipient from EPFL, 2023**
 - Fellowship offered to the top 3% of applicants for a Master Program at EPFL, which I rejected in order to pursue education at ETH.
- **"Best Freshman Prize", 2021**
 - Yearly prize given to the best performing freshmen (roughly 200) of Politecnico di Milano.
- **“DSU (Diritto Scuola Universitario) Scholarship, 2020-2023**
 - Scholarship issued by the government consisting of 7'000 – 10'000 euros/year for students at Politecnico di Milano.
- **ZeroRobotics, 2018-2020**
 - National Winner (Campionato Italiano), International Finalist with EX-AEQUO 3rd place(2018), International 4th place (2019).
 - Challenge hosted by MIT and NASA on the International Space Station.
- **Math Team Olympiad, 2018-2019**
 - National Finalist (top 15), Regional Gold Medal.

SKILLS

- **Software Development**
 - Python/MATLAB for multipurpose tasks
 - CSS/HTML for web development
 - Software version control using Git
- **Multiphysics Simulations**
 - ANSYS package (EDT, HFSS, Q3D) for EM-field, S-parameters, and circuit elements simulations.
 - COMSOL for EM-field simulation
- **RF-circuits design**
 - QUCS for elementary circuit simulation
 - GDSpy for custom on-chip designs
- **Tiramisù Expert**
 - Both at eating and making (no complaints received so far)